Final exam · Unit 5 · Fourier series and convergence · 2.5 points · 50 min · Standard exam level

P20 — Fourier series of the full-wave rectified sine $|\sin x|$: coefficients by product-to-sum formulas, uniform convergence with an error bound, and two numerical series

The output voltage of a full-wave rectifier fed with $\sin x$ *is the* 2π*-periodic function*

$$f(x) = |\sin x|, \qquad -\pi < x \le \pi,$$

extended periodically to $\mathbb{R}$*. Use the convention*

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} \left(a_n \cos nx + b_n \sin nx\right), \qquad a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f\,dx, \quad a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f\cos nx\,dx, \quad b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f\sin nx\,dx.$$

(a) Compute all the Fourier coefficients of f*, writing out the product-to-sum formula you use. Treat the case* $n = 1$ *separately.*

(b) To which function does the series converge, and is the convergence uniform on $\mathbb{R}$*? Justify both answers, and give an* N *such that the partial sum* S_N *(all terms with* $n \le N$*) differs from* f *by less than* 0.035 *everywhere.*

(c) Evaluating the series at suitable points, find

$$S = \sum_{k=1}^{\infty} \frac{1}{4k^2 - 1} \qquad \text{and} \qquad T = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{4k^2 - 1}.$$

Check the value of S *independently with partial fractions.*

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Data: $f(x) = |\sin x|$ on $(-\pi, \pi]$, period $2L = 2\pi$ ($L = \pi$).
- Regularity: f is continuous on $\mathbb{R}$ (also across $x = \pm\pi$, where $|\sin \pi| = 0 = |\sin(-\pi)|$), and piecewise smooth: $f'(x) = \cos x$ on $(0, \pi)$ and $-\cos x$ on $(-\pi, 0)$, with corners (jumps of f') at $x = k\pi$.
- Symmetry: $f(-x) = |-\sin x| = f(x)$, so f is **even**. In fact f has period π, which will kill the odd harmonics.

Phase 2. Canonical formulation

Even f (Haberman Sec. 3.3): $b_n = 0$ and the series is a cosine series with

$$a_0 = \frac{1}{\pi}\int_0^{\pi} f\,dx, \qquad a_n = \frac{2}{\pi}\int_0^{\pi} f\cos nx\,dx,$$

and on $[0, \pi]$, $f(x) = \sin x$.

Part (a) · 1.2 points

Fourier coefficients

Step 1. Sine coefficients

$f(x)\sin nx$ is even times odd, i.e. odd, so its integral over the symmetric interval $(-\pi, \pi)$ vanishes: substituting $x = -s$, $dx = -ds$, $\int_{-\pi}^{0} f(x)\sin nx\,dx = \int_{\pi}^{0} f(s)(-\sin ns)(-ds) = -\int_0^{\pi} f(s)\sin ns\,ds$. Hence $b_n = 0$ for all n.

Step 2. Mean value

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|\sin x|\,dx = \frac{1}{\pi}\int_0^{\pi}\sin x\,dx = \frac{1}{\pi}\big[-\cos x\big]_0^{\pi} = \frac{1}{\pi}\big(-\cos\pi + \cos 0\big) = \frac{2}{\pi}.$$

Step 3. The case $n = 1$

$\sin x\cos x = \frac{1}{2}\sin 2x$, so

$$a_1 = \frac{2}{\pi}\int_0^{\pi}\frac{\sin 2x}{2}\,dx = \frac{1}{\pi}\Big[-\frac{\cos 2x}{2}\Big]_0^{\pi} = \frac{1}{\pi}\Big(-\frac{\cos 2\pi}{2} + \frac{\cos 0}{2}\Big) = 0.$$

Step 4. The case $n \geq 2$

Product-to-sum with $A = x$, $B = nx$: $\sin x \cos nx = \frac{1}{2}\big[\sin(1+n)x + \sin(1-n)x\big] = \frac{1}{2}\big[\sin(n+1)x - \sin(n-1)x\big]$. For an integer $m \geq 1$, with the substitution $s = mx$, $ds = m\,dx$:

$$\int_0^{\pi} \sin mx\,dx = \Big[-\frac{\cos mx}{m}\Big]_0^{\pi} = \frac{1-\cos m\pi}{m} = \frac{1-(-1)^m}{m}.$$

With $m = n+1, 1-(-1)^{n+1} = 1+(-1)^n$; with $m = n-1 \geq 1, 1-(-1)^{n-1} = 1+(-1)^n$. Therefore

$$a_n = \frac{2}{\pi}\cdot\frac{1}{2}\Big[\frac{1+(-1)^n}{n+1} - \frac{1+(-1)^n}{n-1}\Big] = \frac{1+(-1)^n}{\pi}\cdot\frac{(n-1)-(n+1)}{n^2-1} = -\frac{2\big(1+(-1)^n\big)}{\pi(n^2-1)}.$$

- n odd: $1+(-1)^n = 0$, so $a_n = 0$ (consistent with Step 3 and with the period π of f).
- $n = 2k$ even: $1+(-1)^n = 2$ and $n^2-1 = 4k^2-1$, so $a_{2k} = -\dfrac{4}{\pi(4k^2-1)}$.

Step 5. The series

$$|\sin x| \sim \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos 2kx}{4k^2-1}.$$

First terms: $a_2 = -\frac{4}{3\pi}$, $a_4 = -\frac{4}{15\pi}$, $a_6 = -\frac{4}{35\pi}$.

KEY RESULTS

- $b_n = 0$; $a_0 = \dfrac{2}{\pi}$; $a_1 = 0$; $a_n = 0$ for odd $n \geq 3$; $a_{2k} = -\dfrac{4}{\pi(4k^2-1)}$.
- $|\sin x| = \dfrac{2}{\pi} - \dfrac{4}{\pi}\displaystyle\sum_{k=1}^{\infty}\frac{\cos 2kx}{4k^2-1} = \frac{2}{\pi} - \frac{4}{\pi}\Big(\frac{\cos 2x}{3} + \frac{\cos 4x}{15} + \frac{\cos 6x}{35} + \cdots\Big)$.

Part (b) · 0.6 points

Convergence

Step 1. Pointwise limit

f is piecewise smooth on $[-\pi,\pi]$, so by the Fourier convergence theorem (Haberman Sec. 3.2) its series converges at each x to $\frac{1}{2}\left[f(x^+)+f(x^-)\right]$ of the periodic extension. The periodic extension of $|\sin x|$ is $|\sin x|$ itself on all of $\mathbb{R}$, which is continuous, so the average equals $f(x)$: the series converges to $|\sin x|$ everywhere, including the corners $x=k\pi$. There is no Gibbs overshoot (Haberman Sec. 3.3: a continuous, piecewise smooth periodic extension gives a continuous limit).

Step 2. Uniform convergence

For all x, $\left|\dfrac{\cos 2kx}{4k^2-1}\right| \le \dfrac{1}{4k^2-1} = M_k$ and $\sum M_k < \infty$ (it behaves like $\sum \frac{1}{4k^2}$). By the Weierstrass M-test the series converges absolutely and uniformly on $\mathbb{R}$ (equivalently, Haberman Sec. 3.4: a continuous periodic extension with piecewise smooth derivative has a uniformly convergent Fourier series).

Step 3. Error bound

Partial fractions: $\dfrac{1}{4k^2-1} = \dfrac{1}{(2k-1)(2k+1)} = \dfrac{1}{2}\left(\dfrac{1}{2k-1}-\dfrac{1}{2k+1}\right)$ (check: $\frac{1}{2}\cdot\frac{(2k+1)-(2k-1)}{(2k-1)(2k+1)} = \frac{1}{4k^2-1}$). The tail telescopes:

$$\sum_{k=K+1}^{\infty}\frac{1}{4k^2-1} = \frac{1}{2}\left[\left(\frac{1}{2K+1}-\frac{1}{2K+3}\right)+\left(\frac{1}{2K+3}-\frac{1}{2K+5}\right)+\cdots\right] = \frac{1}{2(2K+1)}.$$

The partial sum with all harmonics $n \le N = 2K$ keeps the terms $k \le K$, so

$$|f(x)-S_{2K}(x)| \le \frac{4}{\pi}\sum_{k>K}\frac{1}{4k^2-1} = \frac{2}{\pi(2K+1)} \quad \text{for all } x.$$

$\frac{2}{\pi(2K+1)} < 0.035 \iff 2K+1 > \frac{2}{0.035\pi} = 18.19 \iff K \ge 9$. $K=9$ ($N=18$) already gives $\frac{2}{19\pi} = 0.0335$; with $K=10$ ($N=20$) the bound is $\frac{2}{21\pi} \approx 0.0303$. Any $N \ge 18$ works. The bound is attained at $x=0$, where every cosine equals 1.

KEY RESULTS

- The series converges to $|\sin x|$ at **every** $x \in \mathbb{R}$ (no jumps), and the convergence is uniform.
- $\sup_x|f-S_{2N}| \le \dfrac{4}{\pi}\sum_{k>N}\dfrac{1}{4k^2-1} = \dfrac{2}{\pi(2N+1)}$; any $N \ge 18$ works; $N=18$ (terms up to $\cos 18x$) gives $\le \dfrac{2}{19\pi} \approx 0.0335 < 0.035$.

Part (c) · 0.7 points

Two numerical series

Step 1. At $x = 0$

By (b) the series equals $f(0) = |\sin 0| = 0$ there:

$$0 = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{1}{4k^2-1} \implies S = \frac{2/\pi}{4/\pi} = \frac{1}{2}.$$

Step 2. At $x = \pi/2$

$f(\pi/2) = 1$ and $\cos(k\pi) = (-1)^k$:

$$1 = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^k}{4k^2-1} = \frac{2}{\pi} + \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{4k^2-1} = \frac{2}{\pi} + \frac{4}{\pi}T.$$

So $T = \dfrac{\pi}{4}\left(1 - \dfrac{2}{\pi}\right) = \dfrac{\pi-2}{4} \approx 0.2854$. Check with the first terms: $\frac{1}{3} - \frac{1}{15} + \frac{1}{35} - \frac{1}{63} = 0.2794$, approaching 0.2854 with alternating error smaller than the next term $\frac{1}{99}$.

Step 3. Independent check of S

From the telescoping of (b), Step 3, with $K = 0$: the partial sum is $S_K = \sum_{k=1}^{K}\frac{1}{4k^2-1} = \frac{1}{2}\left(1 - \frac{1}{2K+1}\right) = \frac{1}{2} - \frac{1}{2(2K+1)}$, so $S = \lim_{K\to\infty} S_K = \frac{1}{2}$, the same value as in Step 1.

KEY RESULTS

- $x = 0$: $S = \dfrac{1}{2}$.
- $x = \pi/2$: $T = \dfrac{\pi-2}{4} \approx 0.2854$.
- Partial sums $S_K = \dfrac{1}{2} - \dfrac{1}{2(2K+1)} \to \dfrac{1}{2}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** the coefficients decay like $1/n^2$, the rate expected for a continuous function with a jump in its derivative (a function with jumps would give $1/n$); the mean value $a_0 = 2/\pi \approx 0.637$ is the DC output of the rectifier, the classical result for a full-wave rectifier of unit amplitude.

- **Checks:** every coefficient by direct integration over $(-\pi, 0)$ and $(0, \pi)$ (script, $n \le 11$); the series at $x = 0.7$ and $x = -2.5$ matches $|\sin x|$; Parseval's identity (extension: $\frac{1}{\pi}\int_{-\pi}^{\pi} f^2 dx = 2a_0^2 + \sum a_n^2$ in this convention) gives $\sum \frac{1}{(4k^2-1)^2} = \frac{\pi^2-8}{16}$, also checked. Term-by-term differentiation (Haberman Sec. 3.4) is allowed because f is continuous and periodic: the derived series $\frac{8}{\pi}\sum \frac{k \sin 2kx}{4k^2-1}$ converges to f', and to 0 (the average of the jump ± 1) at $x = k\pi$.
- **Common mistakes in the exam:** (1) using the formula for a_n with $n = 1$, which divides by $n - 1 = 0$; (2) integrating $\sin x \cos nx$ over $(-\pi, \pi)$ without the absolute value, which gives $a_n = 0$ for all n; (3) mixing conventions: with $\frac{a_0}{2}$ in front, $a_0 = \frac{4}{\pi}$, not $\frac{2}{\pi}$; (4) claiming only "convergence in the mean" or a Gibbs overshoot at $x = 0$: the extension is continuous, so convergence is uniform.